

Unit III
MEMORY MANAGEMENT

CourseObjective–Co3: To interpret various memory management schemes.

CourseOutcome–CO3: Familiarize in memory management and storage management techniques

Part -A

S.No.	Question	BTL*																		
1.	Recall paging and list its advantages. Paging is a memory management scheme used by operating systems to efficiently manage and allocate physical memory. It eliminates fragmentation by dividing the process's virtual memory into fixed-sized blocks called pages and mapping them to equally sized frames in physical memory. The operating system maintains a page table to keep track of which virtual pages are mapped to which physical frames Advantages of Paging 1. Eliminates External Fragmentation 2. Efficient Memory Utilization 3. Simplifies Memory Allocation	K1																		
2.	List two differences between logical and physical addresses. <table><tr><th>Logical address</th><th>Physical address</th></tr><tr><td>An address generated by CPU is referred to us a logical address</td><td>An address seen by memory unit that is, the one loaded into the memory address register of the memory is referred to as physical address.</td></tr><tr><td>The set of all logical address generated by a program is a logical address space</td><td>The set of all physical address corresponding to these logical addresses is a physical address</td></tr><tr><td>For user view.</td><td>For system view.</td></tr><tr><td>The user program deals with logical address or these are generated by user (program).</td><td>These are generated by memory management unit (MMU).</td></tr></table>	Logical address	Physical address	An address generated by CPU is referred to us a logical address	An address seen by memory unit that is, the one loaded into the memory address register of the memory is referred to as physical address.	The set of all logical address generated by a program is a logical address space	The set of all physical address corresponding to these logical addresses is a physical address	For user view.	For system view.	The user program deals with logical address or these are generated by user (program).	These are generated by memory management unit (MMU).	K2								
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3.	Differentiate paging and segmentation. <table><tr><th>Feature</th><th>Paging</th><th>Segmentation</th></tr><tr><td>Definition</td><td>A memory management scheme that divides the process into fixed-sized blocks called pages.</td><td>A memory management scheme that divides the process into variable-sized segments based on logical divisions.</td></tr><tr><td>Size</td><td>Pages are of fixed size.</td><td>Segments are of variable size depending on the program's structure.</td></tr><tr><td>Fragmentation</td><td>Eliminates external fragmentation but may cause internal fragmentation.</td><td>Eliminates internal fragmentation but may cause external fragmentation.</td></tr><tr><td>Addressing</td><td>Uses a page table to map logical pages to physical frames.</td><td>Uses a segment table to map logical segments to physical memory.</td></tr><tr><td>Logical Division</td><td>Divides memory purely based on size, without considering the program's structure.</td><td>Divides memory based on logical divisions such as functions, arrays, stacks, etc.</td></tr></table>	Feature	Paging	Segmentation	Definition	A memory management scheme that divides the process into fixed-sized blocks called pages.	A memory management scheme that divides the process into variable-sized segments based on logical divisions.	Size	Pages are of fixed size.	Segments are of variable size depending on the program's structure.	Fragmentation	Eliminates external fragmentation but may cause internal fragmentation.	Eliminates internal fragmentation but may cause external fragmentation.	Addressing	Uses a page table to map logical pages to physical frames.	Uses a segment table to map logical segments to physical memory.	Logical Division	Divides memory purely based on size, without considering the program's structure.	Divides memory based on logical divisions such as functions, arrays, stacks, etc.	K2
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4.	Write the purpose of paging. The purpose of paging is to reduce memory overhead by dividing a large page table into smaller, hierarchical levels. This improves memory efficiency, supports large address spaces, and ensures that only necessary portions of the page table are loaded, minimizing wasted memory.	K1
5.	Recall the working set model. The Working Set Model is a memory management concept that defines the set of pages a process actively uses during a given time window (Δ). It helps in minimizing page faults by keeping frequently accessed pages in memory. The OS uses this model to allocate an appropriate number of frames to a process, improving efficiency and reducing thrashing .	K1
6.	In memory management consider the program named as Stack1 which size is 100 KB. This program is loaded in the main memory from 2100 to 2200KB. Show the contents of the page map table for the given scenario. Step 1: Calculate the Number of Pages Since Program Size = 100 KB and Page Size = 10 KB , the number of pages required: $100\text{KB} / 10\text{KB} = 10\text{Pages}$. Step 2: Page Mapping The starting address is 2100 KB , and the pages are stored sequentially in memory. 0 2100 KB 1 2110 KB 2 2120 KB 3 2130 KB 4 2140 KB 5 2150 KB 6 2160 KB 7 2170 KB 8 2180 KB 9 2190 KB	K2
7.	Name the counting-based page replacement algorithms. Counting-based page replacement algorithms use counters to track page usage and make decisions on which page to replace when a new page needs to be loaded into memory. The two main counting-based algorithms are: Least Frequently Used (LFU) Most Frequently Used (MFU)	K1
8.	Recall demand paging. Demand paging is a memory management technique where pages are loaded into memory only when they are needed during program execution, rather than preloading the entire program. If a required page is not in memory, a page fault occurs, and the operating system loads the missing page from disk into RAM.	K1
9.	State Belady's anomaly. Belady's Anomaly is a phenomenon in FIFO (First-In-First-Out) page replacement where increasing the number of allocated page frames unexpectedly increases the number of page faults instead of reducing them. This contradicts the general expectation that adding more memory should improve performance.	K1

10.	Mention the significance of LDT and GDT in segmentation. Global Descriptor Table (GDT) – Stores system-wide segment descriptors (kernel code, shared memory). Local Descriptor Table (LDT) – Stores process-specific segments (user code, data), ensuring process isolation.	K2
11.	Outline the steps involved in handling a page fault in demand paging. Demand paging is a memory management technique where pages are loaded into memory only when required during execution, instead of preloading the entire program. If a referenced page is not in memory, a page fault occurs, and the operating system loads the missing page from disk into RAM. • Page Fault Occurs – The CPU tries to access a page not in memory. • Trap to OS – The operating system detects the page fault and pauses the process. • Check Validity – OS checks if the page is valid and available on disk. • Find a Free Frame – If available, a free frame is selected in memory. If not, a page replacement algorithm is used to free up space. • Load Page from Disk – The required page is fetched from disk into the allocated frame. • Update Page Table – The page table is updated with the new frame number. • Restart Process – The process resumes execution from the interrupted instruction.	K2
12.	Recall swapping. Swapping is a memory management technique where an entire process is temporarily moved from main memory (RAM) to secondary storage (disk) to free up space for other processes. It is later brought back into memory for execution.	K1
13.	State Locality of reference. Locality of reference is the tendency of a process to access the same memory locations (temporal locality) or nearby locations (spatial locality) frequently. In an operating system, it helps optimize caching, paging, and process scheduling by keeping frequently used data in faster memory, reducing page faults and improving performance.	K1
14.	Mention the purpose of Paging. • Paging reduces memory overhead by dividing a large page table into smaller hierarchical tables. • This helps manage large address spaces efficiently and minimizes wasted memory. • Paging improves address translation by loading only necessary parts of the page table into memory.	K2
15.	Recall External Fragmentation. External fragmentation occurs when free memory is divided into small, non-contiguous blocks, making it difficult to allocate large contiguous memory spaces to processes, even though enough total free memory exists. It happens in dynamic memory allocation systems like paging and segmentation.	K3

16.	Consider a logical address space of 64 pages of 1024 words each, mapped onto a physical memory of 32 frames. a. How many bits are there in the logical address? b. How many bits are there in the physical address? a. Logical address: 16 bits b. Physical address: 15 bits	K3
17.	State thrashing. Thrashing occurs when a system spends excessive time swapping pages between RAM and disk instead of executing processes, leading to poor performance. It happens when too many processes compete for limited memory, causing frequent page faults. Solutions include working set model and efficient page replacement algorithms like LRU to reduce unnecessary swaps.	K1
18.	Evaluate the maximum number of pages needed if a system supports 16 bit address line and 1K page size. Given Information: Address bus size: 16-bit (This means the logical address space = 2^{16} bytes = 65,536 bytes) Page size: 1 KB = 2^{10} bytes = 1024 bytes Formula Used: Number of Pages = Logical Address Space Size / Page Size Calculation: Number of Pages = 2^{16} bytes / 2^{10} bytes/page = $2^{16-10} = 2^6 = 64$ pages	K4
19.	State address binding and list the stages at which it occurs. Address binding is the process of mapping program addresses to physical memory addresses. It can occur at three stages: Compile-time binding – Absolute addresses are assigned if the memory location is known. Load-time binding – Relocatable addresses are assigned and converted to absolute addresses during loading. Execution-time binding – Logical addresses are translated to physical addresses dynamically during execution.	K2
20.	Recall compaction and list the situations in which it is applied. Compaction is a memory management technique used to reduce external fragmentation by rearranging memory contents to create a single large block of free space. It is applied in the following situations - When external fragmentation occurs, preventing allocation of large memory blocks. - In dynamic memory allocation, when processes terminate and leave scattered free memory gaps. - Used in contiguous memory allocation but not in paging, since paging eliminates fragmentation.	K1

21.	State Virtual Memory. Virtual memory is a memory management technique that uses disk space as an extension of RAM , allowing large programs to run even if physical memory is limited. It works through paging or segmentation , enabling efficient multitasking and reducing memory constraints.	K1
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Part B

S.No.	Question	BTL *
1.	Explain the various memory management techniques in operating systems. Compare and contrast contiguous memory allocation and non-contiguous memory allocation.	K4
2.	Explain how paging supports virtual memory with neat sketch. Explain how logical address is translated into physical address using paging mechanism. (16)	K2
3.	Describe segmentation with a neat diagram. How does segmentation solve fragmentation issues and explain segmentation with paging.	K2
4.	Discuss about following page replacement policy with neat sketch: (i) FIFO (ii) LIFO (iii) LRU (iv) Optimal. Consider a virtual page reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1. Suppose a demand paged virtual memory system running on a computer system such that the main memory has 3 page frames. Which one has minimum number of page faults? (16) (OR) Explain Page Replacement Technique with an example. (16)	K3 K2
5.	Explain virtual memory and demand paging in detail. How does demand paging help improve system performance? Discuss page faults and their handling.	K2
6.	Describe contiguous memory allocation techniques. Discuss the problems of fragmentation and solutions like compaction.	K2
7.	Explain the concept of thrashing in memory management. How does working set model help in controlling thrashing?	K2
8.	Outline swapping in memory management. Explain how swapping improves memory utilization and its advantages and disadvantages.	K2
9.	Case Study: "StudyTab" Tablet Memory Problems Scenario: A company makes cheap tablets for students with only 2GB of RAM. Students need to use multiple apps at once - like a video lesson app, a coding practice app, a textbook reader, and a quiz app. Problems: 1. Memory Waste: The tablet's memory gets divided into small unusable pieces between apps 2. Big App Trouble: Some educational apps need 1.5GB, but won't run alongside other apps even though there's total memory available 3. Extreme Slowness: When switching between apps, the tablet freezes and shows "loading" for a long time	K2

Current System:

- 2GB physical memory
- Uses basic memory allocation that looks for continuous free space
- No virtual memory system
- 4KB page size (when they try paging)

Questions:**a. Memory Management Improvement (8)**

- (i) Explain why their current memory system creates wasted space between apps
- (ii) Describe how switching to a paging system would fix this problem
- (iii) Calculate: If they use 4KB pages with 2GB RAM, how many total pages exist?
- (iv) Explain simply why a two-level page table saves memory space

b. Solving Performance Issues (8)

- (i) Explain in simple terms what happens when the tablet shows "excessive page faults"
- (ii) Explain why running video + coding apps causes it
- (iii) Compare FIFO and LRU page replacement: which is better for student tablets and why?
- (iv) Describe how "Copy-on-Write" helps when students make copies of their coding projects